# KEPLER

## User Guide — Installation

## Installation

Kepler requires Python 3.8 or later and five supporting libraries (NumPy, Pillow, SciPy, tifffile, scikit-image). The included installer handles everything automatically. Choose the section below for your operating system.

### Installing on Windows 11

Kepler runs on Windows 11. The installer checks for Python and all required libraries automatically — if anything is missing it installs it for you. No programming knowledge is needed.

#### What you need before you start

- A PC running Windows 11 (Home, Pro, or Education).
- An internet connection for the initial installation (to download Python and libraries if needed).
- About 150 MB of free disk space in the worst case — less if Python is already installed.
- Administrator rights are recommended but not required. Right-clicking install.bat and choosing Run as administrator gives the smoothest experience.

#### Step-by-step installation

**1 Extract the Kepler zip file**

Right-click the Kepler zip file you downloaded and choose Extract All. Extract it to a convenient location such as your Desktop or Downloads folder. You will see a folder containing install.bat and several other files.

**2 Right-click install.bat and run as administrator**

Inside the extracted folder, right-click install.bat and choose Run as administrator.

**Important:**
Windows may show a SmartScreen warning: "Windows protected your PC". This appears because Kepler is newly released, independent software not yet in Microsoft's database. does not mean the software is harmful.

To proceed:

1. Click More info  (the small gray link under the warning text)
2. Click Run anyway  (a button that now appears)

A terminal window will briefly appear, then the Kepler Installer window will open.

## 3 Welcome page — click Begin

The Welcome page shows a list of what the installer will do. Click Begin ▶ to continue.

**Tip:**
If a yellow box says NOT RUNNING AS ADMINISTRATOR, the installer will still work but the system-wide PATH update may be skipped. You can re-run as administrator later if needed.

## 4 Warnings page — read and click I Understand

This page lists three potential prompts you may see during installation:

**UAC prompt ("Allow this app to make changes?"):** Click Yes. This appears only if Python needs to be installed.

**Python installer checkbox:** If the Python installer opens, look for a checkbox at the very bottom of the first screen that says Add python.exe to PATH. Tick it before clicking anything else. This is critical — without it Kepler will not launch from shortcuts or the terminal.

**Antivirus prompt:** If your antivirus flags pip.exe, choose Allow. This is a known false positive — pip is Python's official package downloader.

Click I Understand ▶ to continue.

## 5 System Check page — wait for results, then click Continue

The installer scans your system. You will see a checklist of items:

- Green ticks (✓) — already installed, nothing to do.
- Orange items — missing, will be installed automatically.

Wait for all items to resolve. When the Continue ▶ button becomes active, click it.

**Tip:**

If Python is shown as Not found, the installer will download and install it automatically. You will see a UAC prompt and possibly the Python installer window — follow the instructions in Step 4 above.

## 6 Installation Options page — review and click Install

Review the installation folder. The default is:

```
C:\Users\YourName\AppData\Local\Kepler
```

You can change this by clicking Browse… if you prefer a different location. Leave the checkboxes ticked:

- Create Desktop shortcut — adds a Kepler icon to your Desktop.
- Add Kepler to the Start Menu — adds Kepler to All Programs.
- Add Kepler to system PATH — lets you type kepler in any terminal.

Click Install ▶ to begin.

## 7 Installation progress — wait for completion

A progress bar and log show what is happening. The installer will:

- Install or update Python if needed.
- Install NumPy, Pillow, SciPy, tifffile, and scikit-image using pip.
- Copy the Kepler application files to the installation folder.
- Create the Desktop shortcut and Start Menu entry.

Typical time: under 2 minutes if Python is already installed, 5–10 minutes if Python needs to be downloaded.

## 8 Done — click Finish or Launch Kepler

When the installer shows Kepler Successfully Installed!, you are done.

✓ Kepler has been installed.
You can now launch it from:

- The Kepler shortcut on your Desktop
- Start › All Programs › Kepler
- Any terminal — type: kepler

Tick Launch Kepler when I click Finish if you want to open Kepler immediately, then click Finish.

### Troubleshooting — Windows

| Problem | Solution |
|---|---|
| **Desktop icon has no image (blank icon)** | Run install.bat again as administrator. The icon file was not found at shortcut creation time. |
| **"Python was not found" when double-clicking Kepler** | Python's PATH was not set. Re-run the Python installer, tick Add python.exe to PATH, then re-run install.bat. |
| **SmartScreen blocks install.bat** | Click More info, then Run anyway. See Step 2 above. |
| **Installer says "pip is not available"** | Open a terminal and run: python -m ensurepip --upgrade, then re-run install.bat. |
| **Kepler opens a black window and closes** | A library may be missing. Open a terminal and run: pip install numpy Pillow scipy tifffile scikit-image |
| **To uninstall Kepler** | Open the installation folder (C:\Users\YourName\AppData\Local\Kepler by default) and run uninstall.bat. This removes Kepler files but leaves Python and its libraries in place. |

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## Installing on Linux Mint

Kepler runs on Linux Mint (and most Ubuntu-based distributions). The installer handles Python library installation automatically using apt and pip, and creates an application menu entry so Kepler appears in your applications like any other program.

### What you need before you start

- Linux Mint 20, 21, or 22 (any edition — Cinnamon, MATE, Xfce).
- Python 3.8 or later — this is already included in all recent Linux Mint versions.
- An internet connection for the initial installation.
- About 65 MB of free disk space in the worst case.
- Your user account password (sudo access) — needed only to install the tkinter GUI library if it is missing. Most systems already have it.

### Step-by-step installation

**1** **Extract the Kepler zip file**

Right-click the Kepler zip file you downloaded and choose Extract Here, or Extract To and pick a location such as your home folder. You will see a folder containing install.sh and several other files.

## 2 Open a terminal in the Kepler folder

Open the Files app (Nemo), navigate into the extracted Kepler folder, then right-click on an empty area of the folder and choose Open Terminal Here. A terminal window will open, already pointed at the right location.

**Tip:**
Alternatively, open the Terminal app from the application menu (Menu → Terminal) and then type: cd ~/Downloads/Kepler (or wherever you extracted the files)

## 3 Run the installer script

In the terminal, type the following command exactly and press Enter:

```
bash install.sh
```

The Kepler Installer window will open after a moment.

**Important:**
If the terminal says Permission denied, type the following first and then try again:

```
chmod +x install.sh
bash install.sh
```

## 4 Welcome page — click Begin

The Welcome page shows what the installer will do. It will have already detected your Linux Mint version. Click Begin ▶ to continue.

## 5 Permissions page — read and click I Understand

This page explains what requires your password and what does not:

**Requires your password (sudo):** installing python3-tk (the GUI toolkit) and python3-pip if they are missing. The terminal where you ran install.sh will show the password prompt — switch to it if the installer appears to be waiting.

**Does not require your password:** installing NumPy, Pillow, SciPy, tifffile, and scikit-image (these go into your user account only, using pip --user), copying the Kepler files, and creating the application menu entry.

Click I Understand ▶ to continue.

**Tip:**
If a green box says sudo is available and cached, the installer will handle everything automatically without pausing for a password.

## 6 System Check page — wait for results, then click Continue

The installer scans your system. You will see a checklist:

- Green ticks (✓) — already installed.
- Orange items (Missing — will install) — the installer will handle these automatically.

When all items resolve and the Continue ▶ button becomes active, click it.

**Important:**
If python3-tk shows as Missing, the installer needs to run: sudo apt install python3-tk

You will be prompted for your password in the terminal window. Switch to the terminal, ty your password, and press Enter. Then come back to the installer window.

## 7 Installation Options page — review and click Install

Review the installation folder. The default is:

```
/home/yourname/.local/share/kepler
```

You can change this by clicking Browse… if you prefer a different location. Leave the checkboxes ticked:

- Create application menu entry — adds Kepler to Menu → Graphics (or similar).
- Create 'kepler' command — lets you type kepler in any terminal.
- Add ~/.local/bin to PATH — ensures the kepler command is always found.

Click Install ▶ to begin.

## 8 Installation progress — wait for completion

A progress bar and scrolling log show what is happening. The installer will:

- Install any missing system packages via apt (may prompt for your password in the terminal).
- Install NumPy, Pillow, SciPy, tifffile, and scikit-image into your user account using pip --user.
- Copy the Kepler application files to the installation folder.
- Create the application menu entry (.desktop file) and the kepler terminal command.

Typical time: under 2 minutes if all libraries are already installed, 3–5 minutes if pip needs to download them.

## 9 Done — click Finish or Launch Kepler

When the installer shows Kepler Successfully Installed!, you are done.

✓ Kepler has been installed.
You can now launch it from:

- Menu → Graphics → Kepler (may need to log out and back in once)
- Any terminal — type: kepler
- Direct: python3 ~/.local/share/kepler/main.py

Tick Launch Kepler now and click Finish to open Kepler immediately.

**Tip:**
If the application menu entry does not appear immediately, log out and log back in. Linux Mint refreshes the application menu on login.

## Troubleshooting — Linux Mint

| Problem | Solution |
|---|---|
| **"kepler: command not found" in terminal** | Run: source ~/.bashrc then try again. Or open a new terminal window. The PATH update takes effect in new sessions. |
| **Installer window does not open (no GUI)** | tkinter may be missing. In the terminal run: sudo apt install python3-tk then run: bash install.sh again. |
| **"Permission denied" when running install.sh** | Run: chmod +x install.sh then: bash install.sh |
| **Kepler appears in the menu but does not open** | Try launching from the terminal: kepler — any error message will be visible there. |

| Problem | Solution |
| --- | --- |
| **"No module named 'numpy'" or similar error** | Run: pip3 install numpy Pillow scipy tifffile scikit-image --user then try launching Kepler again. |
| **To uninstall Kepler** | Open a terminal and run: bash ~/.local/share/kepler/uninstall.sh — this removes the Kepler files, menu entry, and kepler command. Python and its libraries are left in place. |